



Innovative Applications of O.R.

A jump–diffusion Stackelberg stochastic differential game in optimal carbon abatement strategies with green subsidy[☆]

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ABSTRACT

The increasingly severe challenges of global warming necessitate a low-carbon transition that involves quantified estimation of carbon emission reduction policies. Carbon trading with abatement subsidization has proven to be an effective measure for evaluating the social cost of carbon emissions. However, conflicting targets between the government and firms can cause principal–agent issues in reducing carbon dioxide emissions. We propose a novel Itô–Lévy jump–diffusion state equation to depict the dramatic fluctuations of carbon prices, a critical factor in carbon trading. To analyze the complex game behaviors in carbon abatement, we construct a multiple players Stackelberg stochastic differential game model applying stochastic optimal control theory in principal–agent situations. Our model examines the incentive compatibility mechanism while considering the complex behaviors of carbon trading participants with asymmetric information. Optimal feedback control strategies are explicitly presented for the government and firms to achieve their separate targets of maximizing social welfare and production profits. Green subsidization coupled with carbon quotas trading is a viable option for promoting carbon abatement. However, high-carbon firms may attempt to deceive the government by pretending to be low-carbon ones. We illustrate that carbon traders with deceptive intentions lead to negative social welfare and hinder carbon abatement achievements.

1. Introduction

Driven predominantly by carbon emissions, global warming has emerged as a pressing environmental and societal issue (Zhang, Wong, & Klassen, 2024). According to International Energy Agency (2024), China's carbon emissions surged by approximately 565 million tons, reaching 12.6 billion tons. This catapults China to the forefront of global carbon emitters, responsible for a significant 35% of the total emissions globally. With the goal of environmental sustainability, carbon emission reduction policies emerge as a powerful tool. The government incentivizes firms to increase carbon reduction inputs through abatement subsidies, sparking a green transformation in industrial practices. Firms with carbon constraints receive initial emission quotas, kickstarting their participation in the carbon trading market. This strategic allocation sets the stage for a vibrant exchange system, driving responsible carbon emission management. Encouraged to align their emissions with allocated quotas, high-emission enterprises face financial pressures that motivate them to seek innovative ways to diminish their carbon footprint (Wei et al., 2023).

Government–enterprise interactions through carbon emission reduction policies are crucial for achieving carbon neutrality goals. The government plays a pivotal role in guiding enterprises to participate in carbon reduction through various policy instruments and incentive measures, including cap-and-trade regulations, carbon taxes, and emission reduction subsidies. These policies collectively incentivize enterprises to develop green products and reduce emissions. Cap-and-trade regulations are a typical policy instrument implemented by the government to constrain emissions from firms (Bartram et al., 2022; Fan et al., 2023; Pun & Ghamat, 2024). Another common policy is the carbon tax (Bai & Ru, 2024; Silva & Magalhães, 2023; Wei et al., 2024), which significantly impacts the relationships among manufacturers (Dou & Choi, 2024). They have explored the effects of uncertain carbon reduction policies on irreversible decision-making among oligopolistic coal electricity firms (Gowrisankaran et al., 2024) and the role of penalty policies in fostering environmental consciousness among competitive firms (Golmohammadi et al., 2024). Government subsidies are also

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critical in promoting green products with lower carbon emissions (Cohen et al., 2016; Jung & Feng, 2020; Yang et al., 2021). Research has investigated dynamic subsidies in green technologies and their influence on firm decision-making (Chemama et al., 2019). A game-theoretic approach has been used to analyze how environmental subsidies affect manufacturers' incentives to invest in emission-reducing technologies, considering consumer environmental concerns (Bian et al., 2020). Additionally, a general functional form for regulations has been proposed to create a comprehensive decision support system for social planners to improve firms' environmental performance (Pazoki & Zaccour, 2019). Despite these advancements, previous studies have limitations in optimizing the interaction between government policies and corporate behavior in a dynamic and uncertain environment. Our study aims to break the limitation by developing a stochastic differential game framework of jump–diffusion that integrates dynamic policy instruments with corporate decision-making under uncertainties of the jump in carbon prices.

Accurately assessing the social cost of carbon emissions is essential for incentivizing firms to develop green technologies and reduce their carbon footprint. However, the direct implementation of carbon emission reduction policies may be ineffective if they do not account for the complex feedback behaviors of firms. Firms often respond strategically to regulatory measures, and these responses can significantly influence the effectiveness of carbon emission reduction policies. Therefore, understanding the dynamic interactions between governments and enterprises is crucial for designing more effective carbon abatement strategies and evaluating carbon emission reduction effects. The game-theoretical approach offers a valuable framework for analyzing these interactions. It explores how different policy instruments influence corporate behavior and identifies optimal policy designs that align corporate incentives with broader environmental goals (Bai et al., 2022; Franks et al., 2023). The policy-oriented carbon trading market is a prime example of a multifaceted game involving key players such as governments, regulatory departments, and carbon trading firms (Bertinelli et al., 2014; Chung et al., 2013; Ma et al., 2021). By combining game-theoretical models with behavior analysis from the carbon trading market, researchers can gain a deeper understanding of the incentives and constraints faced by various carbon trading participants.

Optimization in a game-theoretical framework is the primary methodology in carbon-constrained multiple players' decision-making. According to Zhou and Wen (2020), carbon reduction targets in operational organizations depend mainly on policy-oriented carbon abatement constraints. The main types in-game modeling can be categorized into static game (Anand & Giraud-Carrier, 2020; Cai et al., 2023; Hafezi et al., 2023; Subramanian et al., 2007), deterministically dynamic game (Martín-Herrán & Rubio, 2021), and stochastic game (Abada et al., 2025; Nkuiya, 2015). Stackelberg game method is widely applied to carbon reduction. Yang et al. (2020) constructed a static Stackelberg game in remanufacturing closed-loop supply chain operating under cap-and-trade regulation. Konur (2017) investigated the equilibrium in a non-collaborative static game constrained by emission targets. Zhang et al. (2020) established a static game on regulatory policy design and selection issues for regulators and manufacturers. Ouardighi et al. (2016) examined the effect of double marginalization on the balance between polluting emissions and abatement activities related to pollution accumulation in a dynamic Stackelberg game. Barbosa-Póvoa et al. (2018) highlighted the need for advanced methods to handle uncertainty and complexity in dynamic optimization problems in sustainable operations management. However, few papers consider the jump–diffusion stochastic differential game in a policy-oriented carbon trading market, especially in a stochastic dynamic Stackelberg game framework. Nadarajah and Secomandi (2023) pointed out that stochastic optimal control can provide exact solutions for certain dynamic optimization problems allowing for the explicit derivation of the optimal dynamic policy. Our study develops a stochastic dynamic

Stackelberg game framework involving the government and firms, deriving an explicit solution for dynamic green subsidies. It effectively resolves the optimal feedback control problem for carbon emission reduction in the context of carbon price jump risk. The existing studies have not yet simultaneously addressed both stochastic elements and dynamic game-theoretic approaches. In particular, we simultaneously characterize the leader–follower game with carbon-constrained targets and the strategic choices of complex behaviors in the principal–agent extension. The application of stochastic optimization methodologies with jump–diffusion processes provides valuable theoretical insights into navigating climate-related economic challenges (Chronopoulos & Lumberras, 2017; Josa-Fombellida & López-Casado, 2023; Siddiqui et al., 2019).

However, there is a research gap in carbon emission reduction policies and game behaviors. An obstacle in designing carbon emission reduction policies is that the government and the firms have conflicting targets. A pivotal challenge in crafting effective carbon emission reduction policies lies in the inherent conflict between the government's objectives of maximizing social welfare and mitigating climate impacts, and firms' pursuit of profit maximization. This disparity underscores the complexity and urgency of developing innovative policy frameworks that can balance these competitive strategies. Our research investigates the interactions between the government and firms and seeks optimal strategies for overcoming the obstacle. This article endeavors to fill the gap by introducing a dynamic carbon pricing model with jump–diffusion characteristics into the optimal emission reduction game. By incorporating this sophisticated pricing mechanism, our research aims to provide a more realistic and comprehensive framework for analyzing the strategic interactions between firms. Our study involves three aspects:

- (1) How to construct the jump–diffusion Stackelberg stochastic differential game in carbon abatement constrained by dynamic carbon price?
- (2) What are the optimal green subsidies for carbon abatement and the optimal feedback control strategies of carbon trading firms?
- (3) How to examine incentive compatibility mechanisms between the government and firms with asymmetric information on the pretending behaviors?

In practice, many countries and regions—including the United States, China, and the European Union—have established carbon markets and introduced subsidy schemes to support carbon reduction efforts. Numerous studies (Chevallier & Sévi, 2014; Koch et al., 2016a; Pan et al., 2022a) have documented that carbon price dynamics often exhibit jump features driven by policy changes or extreme events. Chevallier and Sévi (2014) recommend modeling carbon price dynamics using Lévy processes. Enterprises can broadly be categorized into two types: high carbon-emission enterprises (carbon buyers) and low carbon-emission enterprises (carbon sellers). For example, in the power sector, thermal power producers, especially those relying on coal- or gas-fired generation, are typically classified as high carbon-emission enterprises, whereas electricity companies operating hydropower, wind power, or nuclear facilities are generally considered low carbon-emission enterprises. Under the government's carbon reduction targets and green subsidy policies, these enterprises make production and abatement input decisions to maximize their profits. Because the government's subsidy decisions directly influence enterprise profitability, this interaction naturally gives rise to a game-theoretic problem between the government's abatement subsidy policy and the enterprises' production and abatement strategies. Accordingly, we investigate a Stackelberg stochastic differential game in which carbon price dynamics are characterized using Lévy processes, with the government acting as the leader and production firms as followers. The government (leader) optimizes its green subsidy policy to maximize social welfare, taking into account the enterprises' responses. The enterprises (followers), in turn, make production and abatement input decisions to maximize profits given the government's optimized subsidy scheme.

Importantly, the government provides subsidies only to low carbon-emission enterprises to incentivize carbon reduction. However, this opens the possibility that high carbon-emission enterprises may attempt to misrepresent themselves as low carbon-emission enterprises to obtain subsidies. For instance, some coal-fired power plants or energy-intensive industries have overstated their renewable energy usage, claimed misleading emissions reductions, or manipulated reporting data to secure subsidies intended for genuinely low-carbon or clean energy producers. To capture this phenomenon, we extend our model to a principal–agent framework in which high carbon-emission enterprises may strategically choose whether to disguise themselves, while the government may decide whether to regulate such behavior. The model incorporates both the regulation cost for the government and the penalty imposed on enterprises if such misrepresentation is detected. In this way, our model effectively reflects real-world conditions. The results provide useful policy implications, assisting governments in designing green subsidy schemes that more effectively promote enterprise-level carbon reduction, thereby accelerating the transition toward a green and low-carbon economy.

Our research has several key insights. The corporate disguise tactics enterprises employ do not significantly alter the government's optimal abatement subsidy policy, underscoring the robustness of the policy framework in mitigating potential market distortions. Adopting deceptive strategies to masquerade as low-emission enterprises ultimately obtains higher profit but leads to negative social welfare. Society has to take the consequence of the deceptive behaviors of carbon buyer firms. High-emission enterprises can achieve higher profits by disguising themselves as low-emission enterprises compared to carbon sellers, but at the cost of having to incur additional emissions reduction inputs that are three times those paid by carbon sellers. The carbon trading mechanism crafted within the theoretical model presented in this paper inherently discourages corporate disguise, facilitating a self-correcting market mechanism that promotes honesty and efficiency.

Our contributions are threefold. First, we have developed a theoretical modeling approach for a carbon-constrained stochastic differential game incorporating jump–diffusion characteristics captured by the Lévy process, which reflects dynamic carbon pricing in the policy-driven carbon market. We offer novel insights into government–enterprise interactions, considering the strategic choices of both parties under implicit probabilities and behavioral uncertainties. Firms may disguise their carbon emissions to obtain higher green subsidies, while the government faces challenges regulating firms' behavior due to asymmetric information. This complexity necessitates adjusting firms' optimal feedback strategies, posing technical challenges in solving the coupled Hamilton–Jacobi–Bellman (HJB) equations incorporating jump–diffusion terms.

Second, we underscore the resilience of the policy-driven carbon trading market in real-world industrial applications. This market effectively fosters a significant reduction in overall carbon emissions, devoid of disguising tactics. Conversely, when high-emitting firms employ deceptive practices, carbon emissions escalate, albeit at a progressively diminishing marginal increment. Notably, the optimal abatement subsidy policy addresses the strategic maneuvers of high-emitting enterprises seeking to conceal their emission reduction endeavors, thereby safeguarding the government's optimal distribution of emission reduction subsidies from being disrupted by such disguising strategies.

Third, we propose a theoretical method to evaluate the effectiveness of carbon emission reduction policies through optimal carbon pricing. Our approach demonstrates that green subsidies are an effective measure to promote carbon reduction. This dynamic green subsidy mechanism integrates social welfare and profit maximization objectives, paving the way for an incentive mechanism in carbon reduction and green transition. The optimal green subsidies from the government can motivate enterprises to accelerate their low-carbon transition, while the mechanism also effectively limits total carbon emissions.

The remainder is organized as follows. The literature review is discussed in Section 2. In Section 3, we define the research problem and differentiate the two types of stochastic differential game scenarios. In Sections 4 and 5, we construct a theoretical model with explicit solutions and optimal closed-loop control strategies in basic and principal–agent circumstances, respectively. Numerical simulations are provided in Section 6 to illustrate the methodological applications in carbon emission reduction policies implementation. In Section 7, there are concluding remarks and extended research discussions.

2. Literature review

The existing studies concentrate on the evaluation of carbon emission reduction policies, which can be classified into three streams: the dynamic carbon price with jump–diffusion captured by Lévy processes, the incentive compatibility mechanisms with static or dynamic and stochastic differential game modeling, and the social cost of carbon with carbon emission reduction policies. The distinctions of this paper from the existing literature can be summarized in three aspects: First, we incorporate control variables into the stochastic differential equations that describe carbon prices, thereby integrating carbon price jump risks into the stochastic optimal control model to derive analytical solutions for feedback control strategies. Second, we extend the stochastic differential game to a model constrained by jump–diffusion state equations and introduce complex behaviors of decision-makers into the expanded model. Third, we design a dynamic carbon reduction policy for optimal green subsidies and conduct a precise assessment of the social cost of carbon.

First, researchers concentrate on modeling the dynamic carbon price with jump–diffusion uncertainties. Carbon asset serving as a special financial asset exhibits remarkable jump–diffusion uncertainties including macroeconomic disruptions, carbon emission reduction policies, and technical innovations (Pan et al., 2022b). Flora and Vargiolu (2020) pointed out that the theoretical modeling of dynamic carbon prices has remained an unresolved challenge. In response to the risk of carbon price jumps, much literature focuses on modeling the jump process of dynamic carbon price using compound Poisson processes (Gugler et al., 2023; Olijslagers et al., 2023). Jump risks are the main factors in financial asset pricing captured by Lévy processes (Armerin, 2023; Dungey et al., 2018; Duong & Swanson, 2015; Lee & Hannig, 2010; Nunes & Pimentel, 2017). The stochastic factors are widely applied in modeling the carbon asset pricing (Barnett, 2023; Barnett et al., 2020; Chassagneux et al., 2023; Hong et al., 2023; Ibrahim & Kalaitzoglou, 2016). Regarding option pricing, Kou (2002) and Kou and Wang (2004) have proposed models incorporating double exponential jumps, while mixed-exponential jump models have been explored in Cai and Kou (2011) and Shi (2022).

In contrast to existing literature that uses jump–diffusion processes to analyze dynamic carbon prices, we propose a jump–diffusion stochastic optimal control model that examines carbon price uncertainty from a game-theoretical perspective. Our study incorporates government green subsidies as a control variable within the stochastic differential equation that describes carbon prices characterized by Lévy processes. Using a compound Poisson process to define the magnitude and frequency of price jumps, we have developed a more accurate stochastic differential equation for dynamic carbon prices that includes a jump term. This approach allows us to quantitatively represent carbon price fluctuations, including jump risks, in light of carbon emission reduction policies. We derive the optimal green subsidy by solving the government's social welfare maximization objective function through stochastic optimal control. This establishes the functional relationship between dynamic green subsidies and carbon prices. By integrating control variables into the stochastic differential equation framework, we better understand the uncertainties associated with carbon prices. This improved precision enables the model to better reflect the decision-making behaviors of market participants regarding

carbon price uncertainties. Through simulations of carbon price dynamics under various scenarios, policymakers can effectively evaluate the impact of carbon emission reduction policies and develop more informed and responsive strategies.

Second, plenty of studies concentrate on differential games in operational management constrained by carbon emission reduction policies. Combined carbon emission reduction policies with carbon emission trading is widely applied in the real market. For example, [Yenipazarli \(2016\)](#) investigated the optimal carbon taxation implemented on the manufacturer's and the remanufacturer's decision-making. [Parilina and Zaccour \(2022\)](#) investigated the cooperative strategies in pollution abatement in a dynamic game framework. [Feichtinger et al. \(2022\)](#) proposed a dynamic oligopoly game to analyze the optimal carbon taxation and maximum sustainable yield of firms. Abundant studies focus on incentive compatibility mechanism in carbon reduction with static or deterministically dynamic game models. Incentive compatibility mechanisms ([Danz et al., 2022](#); [Mansour et al., 2020](#); [Moiseva et al., 2017](#); [Witkowski et al., 2023](#)) are widely discussed in the scenarios of inconsistent targets. Carbon abatement is a typical principal–agent problem in the government as a regulator authorizes firms to produce green goods, and firms as agents are responsible for increasing abatement inputs to research and develop abatement technologies with carbon reduction constraints. [Habla and Winkler \(2018\)](#) examined a delegated carbon emissions permit market regulated by carbon emission reduction policies in a non-cooperative game framework. [Chiroleu-Assouline and Lambert-Mogiliansky \(2023\)](#) discussed the firms with abatement efforts incentivized by regulators. As discussed by [Helm and Wirl \(2014\)](#), the principal cooperates with the agent in carbon reduction with strategic options that the willingness to reduce carbon emissions is inapparent information for the principal.

On the flip side, several studies concentrate on static game in carbon reduction ([Drake et al., 2016](#); [Gopalakrishnan et al., 2021](#); [Wang et al., 2021](#); [Xu et al., 2024](#); [Zhang, Zhu, & Lin, 2024](#)). [Bai et al. \(2023\)](#) formulated a decentralized model incorporating energy conservation to analyze the strategic decision of whether to partially integrate some component suppliers utilizing a manufacturer-led static game-theoretic approach. [Long et al. \(2022\)](#) explored the interactions among dominant structural factors, green sensitivity, and consumer preferences regarding green closed-loop supply chains in manufacturing. As discussed in [Lukas and Welling \(2014\)](#), extensive scholarly works on green supply chain management have predominantly centered around addressing the complex issues of environmentally sustainable production. While a limited number of quantitative models have utilized game theory to examine optimal emission reduction strategies, these studies often rely on the assumption of a static carbon price, overlooking the inherent uncertainty and the jump characteristics that capture dynamic carbon asset prices with frequent shocks. Stackelberg game ([Aguad & Thraves, 2024](#); [Crettez et al., 2025](#); [Guzmán et al., 2022](#); [He et al., 2021](#); [Salhab et al., 2022](#)) is applied widely in carbon reduction. [Jiang et al. \(2025\)](#) focused on the supply chain game with carbon footprint reduction and pricing strategies in response to eco-conscious consumers. [Zolfagharinia et al. \(2023\)](#) examined a monopolist firm's decision to design green products ([He et al., 2023](#)) incentivized by subsidies. [Jin et al. \(2022\)](#) examined the influence of government support on the production and green investment decisions of a manufacturing firm facing financing constraints. [Zhuge et al. \(2024\)](#) investigated the game between shipping firms and the regulator constrained by the emission control areas policies. [Yenipazarli \(2016\)](#) analyzed the influence of emissions taxes in shaping the productive optimization and pricing strategies of a manufacturer capable of remanufacturing its products.

This paper enhances the existing literature by extending the Stackelberg game framework between the government and enterprises to a stochastic differential game model that incorporates carbon price jumps. It focuses on the feedback strategies that enterprises adopt in response to government green subsidies. The stochastic dynamic Stackelberg game integrates unpredictable factors in the carbon trading

market, such as the risk of carbon price jumps, alongside policy factors like government green subsidies into the optimization model. This integration increases the model's realism and its ability to reflect market uncertainty as well as the decision-making processes of participants facing uncertainty. Additionally, this stochastic dynamic Stackelberg game can accommodate more complex information structures. For instance, the extended model presented in this paper describes scenarios where game participants possess private information that changes dynamically over time. This adaptability allows the model to more accurately portray the issues of information asymmetry in real-world contexts, illustrating how participants make decisions based on incomplete information. Global warming externalities lead to typical moral hazard problems among firms, as governments permit only them to produce green goods under carbon constraints. In this scenario, private information is crucial in the interactions of the abatement game. Firms cannot predict whether the government will choose to regulate carbon trading, while the government struggles to access reliable information regarding firms' carbon abatement efforts. Effective carbon emission reduction policies can reconcile the conflicting objectives of the game players. Along with a principal–agent extension, the optimal feedback control strategies derived from the stochastic differential game can enhance the effectiveness of carbon emission reduction policies, encouraging firms to pursue a sustainable green transition.

The third stream references the social cost of carbon with carbon emission reduction policies ([Ebadi Torkayesh et al., 2024](#); [Pindyck, 2019](#); [Yu et al., 2022](#); [Zhao et al., 2023](#)). Governments introduce carbon emission reduction policies, like green subsidization, to improve the carbon abatement efforts of firms, while firms take measures to equilibrate their profitability and carbon reduction requirement ([Fu et al., 2023](#)). The effective measures to tackle global warming are to quantifiably evaluate the social cost of carbon emissions and implement environmental policies to compensate for the climatic damages. Precisely pricing carbon emissions formulates the environmental externalities of global warming and mitigates pressures of carbon dioxide emissions by increasing the ignored abatement costs in developing green technologies. The carbon pricing determines the cost of global warming damages and compels high-carbon firms to undertake the abatement obligations. The social cost of carbon can be calculated by estimating environmental externalities with diversified uncertainties ([Eugenia Sanin et al., 2015](#); [Koch et al., 2016b](#)). An approximate tractable expression of the risk-adjusted social cost of carbon is derived by perturbation theory to deal with climatic and economic risk ([van den Bremer & van der Ploeg, 2021](#)). The carbon reduction strategies can be classified as policy-oriented and operation-oriented. Policy-oriented approaches focus on the industrial upgrading constrained by carbon emission reduction policies ([Bai & Ru, 2024](#); [Bian & Zhao, 2020](#); [Gong & Zhou, 2013](#); [Krass et al., 2013](#); [van der Ploeg, 2025](#)). Operation-oriented ones concentrate on supply chain management decision-making with carbon abatement constraint ([Dou & Choi, 2021](#); [Jira & Toffel, 2013](#)). Our research is related to optimal carbon emission reduction policies considering uncertainties, which are widely acknowledged in existing studies with a prominent transformation from the traditional models ([Newbold & Marten, 2014](#); [Pindyck, 2012](#)) to advanced ones ([Cai & Lontzek, 2019](#); [Lontzek et al., 2015](#); [Nordhaus, 2019](#)). [Bigerna et al. \(2023\)](#) investigated the impact of a government subsidy on the likelihood of achieving a policy target and on the resultant welfare, particularly in terms of increasing the proportion of renewable energy production within the overall energy mix. [Li et al. \(2023\)](#) examined government's feed-in tariffs for biomass power generation with carbon reduction goals.

We focus on characterizing the risk of carbon price fluctuations embedded in carbon reduction policies, distinguishing it from the existing literature. Applying the stochastic optimal control theory, we derive an analytical solution for optimal green subsidies. We develop a stochastic differential game between the government and enterprises to

capture the sequential responses of decision-makers to various closed-loop control strategies. Our approach to stochastic dynamic carbon pricing and green subsidies provides regulators and firms with a quantitative evaluation of the social cost of carbon. In our game model, we first examine the non-linear dynamics of green subsidization in the jump–diffusion stochastic differential game framework. We demonstrate that a combination of carbon trading market and optimal green subsidies constitutes an effective carbon emission reduction policy for reducing carbon emissions. High-carbon firms incur additional costs for purchasing carbon quotas, while the green subsidy incentivizes firms to invest in clean technologies and alter their production patterns. Recognizing that carbon reduction is a long-term endeavor requiring ongoing adjustments to relevant policies, the dynamic green subsidies derived in this study present optimal strategies for government action throughout the policy implementation period. These subsidies are designed to encourage enterprises to invest in green technologies and facilitate the low-carbon transition of high-emission manufacturing industries.

3. Problem definition

Two firms supply homogeneous products with Cournot competition on the market. We construct the non-cooperative competition model to characterize the equilibrium abatement inputs and outputs with the stochastic differential state equation of jump–diffusion dynamic carbon price. The two firms have different manufacturing patterns resulting in distinguished actual carbon emissions. The high-carbon firm buys carbon quota from the low-carbon firm in the carbon trading market. Both the carbon quota buyer and seller are regulated by the government. The government provides green subsidies for carbon sellers to encourage firms' participation in green transition. There are two decision-making periods in the non-cooperative Stackelberg stochastic differential game. The first period is when the government determines the optimal green subsidy with social welfare-maximizing goals. The second period is when carbon trading firms optimize the production quantities and abatement inputs with profit-maximizing targets. Firms make operational decisions according to profit-maximizing principles constrained by carbon trading profits and costs. The government as the environmental regulator intends to maximize the social welfare composed of the green utility and the social damage of carbon emissions.

There are two scenarios in game-theoretical models. The first one is the basic model where the carbon quota buyer and seller interact without pretending behaviors. Whether the carbon seller firm can gain green subsidy and whether the carbon buyer pays abatement expenditures are public information. On the contrary, in the principal–agent scenario, the status of firms and the government are private information. The government can choose to regulate carbon traders or not, while firms can choose to strategically overstate their abatement efforts in gaining green subsidies. In the basic game framework, all variables are denoted as $Z^a(t)$ where Z stands for variables in the theoretical analysis sections. Similarly, all variables are denoted as $Z^b(t)$ in the extensional game framework. Two game scenarios will be compared and discussed in Sections 4 and 5, respectively.

4. Basic non-cooperative Stackelberg stochastic differential game

4.1. Model formulation

The stochastically dynamic carbon price is described as a stochastic differential equation with a jump–diffusion term,

$$\begin{cases} dC(t) = \underbrace{[C(t) - \phi s^a(t)]dt}_{\text{Drift term}} + \underbrace{\sigma C(t)dB(t)}_{\text{Diffusion term}} + \underbrace{C(t) \cdot \int_{z \in \mathbb{R}} \gamma(z, z) \tilde{N}(dt, dz)}_{\text{Jump-diffusion term}}, \\ C(0) = C_0, \end{cases} \quad (1)$$

where the compensated Poisson random measure is $\tilde{N}(dt, dz) = N(dt, dz) - \eta(dz)dt$ and $\eta(dz)$ is the Lévy measure of Poisson measure $N(dt, dz)$ where $\eta(dz) = \kappa f(z)dz$ where κ is the jump intensity and $f(z)$ is the probability density. C_0 is a constant that means the initial carbon price at time $t = 0$. $s^a(t)$ denotes the dynamic green subsidy from the government with a positive ratio ϕ impacting the carbon price $C(t)$. If the government implements the green subsidy to incentivize carbon reduction from firms, there are more low-carbon firms that can sell carbon quotas. Then carbon price will decrease owing to more carbon quotas supply.

There are two duopoly firms supplying homogeneous products in the market, similar to Colombo and Labrecciosa (2021). The firm i outputs $q_i^a(t)$ with renewable energy and selling extra carbon quotas, while the firm j outputs $q_j^a(t)$ with fossil energy and buying carbon quotas exceeding the initial quota A . The total output at time t is $Q^a(t) = q_i^a(t) + q_j^a(t)$. The price of homogeneous products is $p^a(t) = \epsilon - Q^a(t)$ where ϵ is the positive pricing parameter. The carbon quotas seller firm i can gain the green subsidy $s^a(t)$ from the government. The instantaneous abatement cost of firm i is $b_i[v_i^a(t)]^2$ where $v_i^a(t)$ is the abatement input and b_i is the positive abatement efficiency parameter (Feichtinger et al., 2022). The production costs of firms i and j are $h_i[q_i^a(t)]^2, h_j[q_j^a(t)]^2$ with positive production parameters h_i, h_j , respectively. $\Pi_i^a(t), \Pi_j^a(t)$ denote the profit functions of carbon trading firms i and j , respectively. The profit of firm i is

$$\Pi_i^a(t) = \underbrace{p(t)q_i^a(t)}_{\text{Production profit}} + \underbrace{C(t)[A - D_i^a(t)]}_{\text{Carbon quotas selling profit}} - \underbrace{b_i[v_i^a(t)]^2}_{\text{Costs}} - \underbrace{h_i[q_i^a(t)]^2}_{\text{Production cost}} + \underbrace{s^a(t)}_{\text{Green subsidy}}, \quad (2)$$

which is composed of production profits, carbon price $C(t)$ multiplies the actual carbon emission $D_i^a(t)$ below the initial carbon quotas as the selling profit, green subsidy $s^a(t)$ and production and abatement costs. The profit of firm j is

$$\Pi_j^a(t) = \underbrace{p^a(t)q_j^a(t)}_{\text{Production profits}} - \underbrace{h_j[q_j^a(t)]^2}_{\text{Production cost}} - \underbrace{C(t)[D_j^a(t) - A]}_{\text{Carbon quotas buying cost}}, \quad (3)$$

where there is no green subsidy for carbon quotas buyer without abatement input and the firm should buy the extra carbon quotas for the actual carbon emission $D_j^a(t)$ excess of the initial carbon quota. The two carbon-trading firms that offer homogeneous products in this model are representative of two types of enterprises: one that adopts green technologies and has low emissions, and the other that uses traditional technologies and has high emissions. In the carbon trading market, there exists carbon market makers who can repurchase excess carbon quotas and replenish insufficient ones (Eugenia Sanin et al., 2015; Koch et al., 2016b). We assume that a carbon trading market maker provides sufficient liquidity to the carbon market, allowing firms i and j to trade carbon quotas freely. The market maker will absorb untraded carbon allowances between the two firms to ensure market balance. To maintain analytical focus, we assume market makers can effectively match the carbon allowances held by enterprises. The government is responsible for issuing carbon allowances and is assumed to allocate an initial amount A to each firm. The focus of this article is on the strategic interactions between the government and enterprises. The government implements green subsidies to guide carbon trading enterprises to reduce carbon emissions. We assume the government allocates carbon quotas equally. This paper does not theoretically model the optimization goals of carbon trading market makers as government-firm intermediaries. A pivotal function of market makers as intermediaries is maintaining strategic inventory reserves, empowering them to provide immediate liquidity during supply–demand mismatches (Bellia et al., 2025; Nimalendran et al., 2024; Spulber, 1996).

The actual carbon emissions of firm i is

$$D_i^a(t) = u_i q_i^a(t) - n_i v_i^a(t), \quad (4)$$

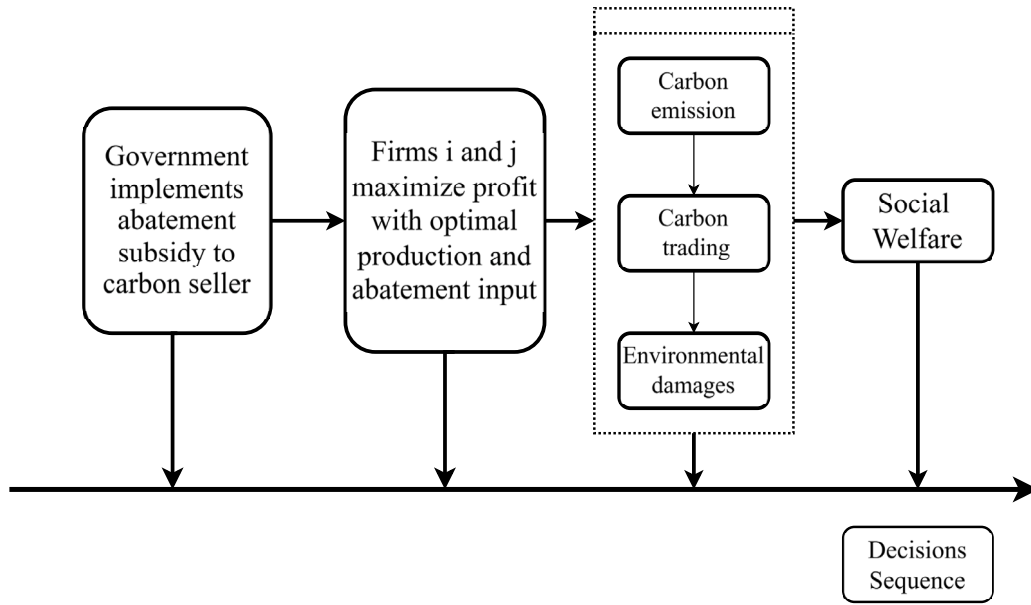


Fig. 1. The sequential decision-making of game players in the basic model.

where n_i, u_i are positive parameters. Eq. (4) describes the relationship between carbon emission $D_i^a(t)$, production $q_i^a(t)$ and abatement input $v_i^a(t)$. If the seller firm increases the abatement input, the carbon emission will decrease, while if the seller increases the production, there will be more carbon emissions. The actual carbon emissions of firm j is

$$D_j^a(t) = u_j q_j^a(t), \quad (5)$$

where u_j is a positive parameter to describe the influential rate of production on carbon emission without abatement input. Since u_i and u_j represent the carbon emissions per unit of production for firms i and j , respectively, the total production $q_i^a(t)$ and $q_j^a(t)$ without carbon abatement input result in total carbon emissions $u_i q_i^a(t)$ and $u_j q_j^a(t)$ for firms i and j , respectively. Since firm i has an abatement input and n_i represents the carbon abatement per unit of input, the total carbon abatement input $v_i^a(t)$ leads to carbon abatement $n_i v_i^a(t)$ for firm i . Thus, the total carbon emission with carbon abatement input for firm i becomes $u_i q_i^a(t) - n_i v_i^a(t)$. Eq. (4) indicates that a firm's actual carbon emissions are directly proportional to its outputs and inversely proportional to the abatement inputs.

The social welfare is composed of green utility $U[s^a(t)]$ minus social damage function which is in direct proportion to total carbon emissions $[D_i^a(t) + D_j^a(t)]$ (Boucekkine et al., 2023; Chen et al., 2024; de Zeeuw, 2008; Tahvonen & Withagen, 1996),

$$W^a(t) = \underbrace{U[s^a(t)]}_{\text{Green utility}} - \underbrace{L[D_i^a(t) + D_j^a(t)]}_{\text{Social damage of total carbon emissions}}, \quad (6)$$

where the constant L is the environmental loss per carbon emission which means the social cost of carbon. The green utility $U[s^a(t)]$ measures the positive part of social welfare $W^a(t)$ derived from green subsidy $s^a(t)$ to encourage firms to increase abatement inputs. Green subsidization increases the green utility $U[s^a(t)]$ including environmental improvements from green technological innovations. Similar to Boucekkine et al. (2023), the green utility is a quadratic function,

$$U[s^a(t)] = a_1 s^a(t) - a_2 [s^a(t)]^2, \quad (7)$$

where a_1, a_2 are positive parameters. The list of notations are displayed in Table D.1 in Appendix D.1.

4.2. Stackelberg stochastic differential game in basic scenario

The Stackelberg stochastic differential game is composed of two sub-games in two decision-making periods. The government is the leader and the production firms are followers. The leader provides optimal green subsidy according to the social welfare maximization considering the followers' responses, while the two firms choose optimal productions and abatement costs based on the leader's optimized subsidy decision to pursue profit-maximizing goals. The followers provide homogeneous products and jointly determine the price described in a Cournot duopoly competition. The goal of this research is to explore a dynamic carbon reduction mechanism in which the government formulates the green subsidy to promote abatement inputs for low-carbon firms that can sell extra carbon quotas. Then the high-carbon firms should buy carbon quotas from the seller and reduce carbon emissions step by step. There are two decision-making periods I and II for the government as the leader and firms as the followers. The sequence of decision-making in the green transition Stackelberg stochastic differential game is presented in Fig. 1.

- The government determines the optimal green subsidy $s^a(t)$ with social welfare maximization goal.
- The carbon trading firms i and j make optimal outputs $q_i^a(t), q_j^a(t)$ with profit maximization objectives.
- The price $p^a(t)$ of products are determined by outputs.
- The carbon quotas seller firm i determines the optimal abatement input $v_i^a(t)$ with given green subsidy $s^a(t)$.
- The firm i sells surplus carbon quotas $[A - D_i^a(t)]$ to carbon trading market and the firm j buys quotas according to extra carbon emissions $[D_j^a(t) - A]$.

Period II The firms determine their productions and abatement inputs with deterministic green subsidy $s^a(t)$ on the goal of profit-maximizing. For the seller firm i , the optimizing objective is the profit function $\Pi_i^a(t)$ in the infinite horizon by choosing the optimal control variables output $q_i^a(t)$ and abatement input $v_i^a(t)$, which is

$$\max_{q_i^a(t), v_i^a(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot \Pi_i^a(t) dt, \quad (8)$$

subject to Eq. (1). Given the information available at time τ , let $c(\tau)$ denote the deterministic value of the stochastically dynamic carbon price $C(t)$ at time τ . Define the deterministic value function $J_i^a[\tau, c(\tau)] =$

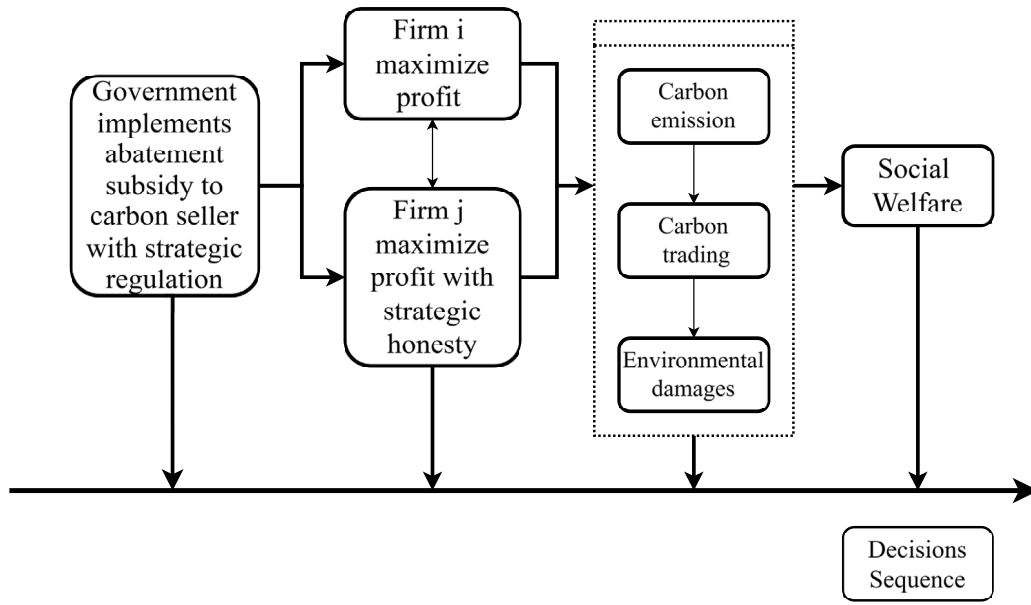


Fig. 2. The sequential decision-making of game players in the extension model.

$\max_{q_i^a(t), v_i^a(t)} E_\tau \int_\tau^{+\infty} e^{-rt} \cdot \Pi_i^a(t) dt$, and $J_i^a[\tau, c(\tau)] = e^{-r\tau} \cdot I_i^a[c(\tau)]$ where $I_i^a[c(\tau)] = k_1 c^2(\tau) + k_2 c(\tau) + k_3$ is a quadratic value function of firm i and k_1, k_2, k_3 are non-zero parameters. e^{-rt} is the discounting factor with a constant discounting rate r . The scripts in the value function $I_i^a[c(\tau)]$ represent the profit in the basic model (the superscript I^a) of firm i . The Hamilton–Jacobi–Bellman (HJB) equation of firm i is

$$\max_{q_i^a(\tau), v_i^a(\tau)} \left[e^{-r\tau} \cdot \Pi_i^a(\tau) + \mathcal{A}_{q_i^a(\tau), v_i^a(\tau)} J_i^a[\tau, c(\tau)] \right] = 0. \quad (9)$$

The generating operator $\mathcal{A}_{q_i^a(\tau), v_i^a(\tau)} J_i^a[\tau, c(\tau)]$ in Eq. (9) is displayed in Equation (C.16) in Appendix C.1.

Substituting Equation (C.16) and Equation (C.17) in Appendix C.1 back to Eq. (9), we have

$$\begin{aligned} \max_{q_i^a(\tau), v_i^a(\tau)} r[k_1 c^2(\tau) + k_2 c(\tau) + k_3] = & \underbrace{[e - q_i^a(\tau) - q_j^a(\tau)] q_i^a(\tau)}_{\text{Production profits}} \\ & + \underbrace{c(\tau) \cdot [A + n v_i^a(\tau) - u_i q_i^a(\tau)] - b_i [v_i^a(\tau)]^2 - h_i [q_i^a(\tau)]^2}_{\text{Carbon position}} + \underbrace{+ s^a(\tau)}_{\text{Costs}} + \underbrace{k_1 \sigma^2 c^2(\tau)}_{\text{Green subsidy}} \\ & + \underbrace{[2k_1 c(\tau) + k_2][c(\tau) - \phi s^a(\tau)]}_{\text{Drift term}} + \underbrace{\int_{z \in \mathbb{R}} k_1 \gamma^2(\tau, z) c^2(\tau) \eta(dz)}_{\text{Jump-diffusion term}}. \end{aligned} \quad (10)$$

We obtain the optimal abatement input and the relationship of the optimal outputs $q_i^a(\tau), q_j^a(\tau)$ for firm i by solving Equation (10),

$$v_i^a(\tau) = \frac{n_i}{2b_i} c(\tau), \quad (11)$$

$$q_i^a \tau = -\frac{u_i}{2h_i + 1} c(\tau) + \frac{\epsilon - q_j^a(\tau)}{2h_i + 1}. \quad (12)$$

Eq. (11) implies that the optimal abatement input is a linear function of deterministic carbon price $c(\tau)$ with positive coefficient $\frac{n_i}{2b_i}$. For the carbon quotas buyer firm j , the optimizing objective is the profit function $\Pi_j^a(t)$ in the infinite horizon by choosing the optimal output $q_j^a(t)$ as the control variable, which is

$$\max_{q_j^a(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot \Pi_j^a(t) dt, \quad (13)$$

subject to Eq. (1). Define $J_j^a[\tau, c(\tau)] = \max_{q_j^a(t)} E_\tau \int_\tau^{+\infty} e^{-rt} \cdot \Pi_j^a(t) dt$, and $J_j^a[\tau, c(\tau)] = e^{-r\tau} \cdot I_j^a[c(\tau)]$ where $I_j^a[c(\tau)] = g_1 c^2(\tau) + g_2 c(\tau) + g_3$

is the quadratic value function of firm j and g_1, g_2, g_3 are non-zero parameters. The HJB equation of the firm i is

$$\max_{q_j^a(\tau)} \left[e^{-r\tau} \cdot \Pi_j^a(\tau) + \mathcal{A}_{q_j^a(\tau)} J_j^a[\tau, c(\tau)] \right] = 0. \quad (14)$$

The HJB equation of firm j becomes

$$\begin{aligned} \max_{q_j^a(\tau)} r[g_1 c^2(\tau) + g_2 c(\tau) + g_3] = & \underbrace{[e - q_i(\tau) - q_j(\tau)] q_j(\tau)}_{\text{Production profits}} - \underbrace{h_j q_j^2(\tau)}_{\text{Production cost}} \\ & - \underbrace{c(\tau) \cdot [u_j q_j(\tau) - A]}_{\text{Carbon position}} \\ & + \underbrace{[2g_1 c(\tau) + g_2][c(\tau) - \phi s^a(\tau)] + g_1 \sigma^2 c^2(\tau)}_{\text{Drift term}} + \underbrace{\int_{z \in \mathbb{R}} g_1 \gamma^2(\tau, z) c^2(\tau) \eta(dz)}_{\text{Jump-diffusion term}}. \end{aligned} \quad (15)$$

We obtain the relationship of the optimal outputs $q_i^a(\tau), q_j^a(\tau)$ for firm j by solving Eqs. (15),

$$q_j^a(\tau) = -\frac{u_j}{2h_j + 1} c(\tau) + \frac{\epsilon - q_i^a(\tau)}{2h_j + 1}. \quad (16)$$

Corollary 1. Solving Eqs. (12) and (16), the optimal productions are

$$\begin{cases} q_i^a(\tau) = \frac{u_j - 2u_i - 2h_j u_i}{4h_i h_j + 4h_i + 4h_j + 3} c(\tau) + \frac{2\epsilon h_j + \epsilon}{4h_i h_j + 4h_i + 4h_j + 3}, \\ q_j^a(\tau) = \frac{u_i - 2u_j - 2h_i u_j}{4h_i h_j + 4h_i + 4h_j + 3} c(\tau) + \frac{2\epsilon h_i + \epsilon}{4h_i h_j + 4h_i + 4h_j + 3}. \end{cases} \quad (17)$$

Eq. (17) indicates that the optimal productions are the linear equation of carbon price $c(\tau)$ with constant coefficients. This optimal linear relationship holds throughout the entire process, which implies the dynamic production $q_i^a(t), q_j^a(t)$ and the stochastically dynamic carbon price $C(t)$ also satisfying the optimal condition Equation (17). This study reveals a quantifiable linear relationship between firms' optimal production strategies and carbon prices subject to jump risks, within the complex market environment shaped by government subsidies and carbon trading mechanisms. This finding offers critical insights for both corporate strategic decision-making and policy design. Traditional management approaches often treat carbon prices as uncontrollable

external costs. In contrast, the model establishes that optimal outputs are directly driven by carbon price dynamics, including jump components. This implies that carbon price management should be integrated into the core of production operations and strategic planning, rather than being relegated solely to compliance or cost-control functions. Executive leadership must monitor, analyze, and forecast carbon price fluctuations and jump risks with the same diligence applied to product market prices and raw material costs, and dynamically adjust production plans in response. Consequently, the role of carbon asset management departments should be elevated to a strategic level. The jump–diffusion process captures abrupt carbon price movements triggered by policy changes, technological breakthroughs, or extreme climate events. The linear relationship between optimal output and carbon prices offers a clear and practical decision rule for firms. Even amid high uncertainty and market shocks, management can swiftly determine the optimal production response based on observed carbon price levels, thereby enhancing both agility and robustness in decision-making. Furthermore, the model incorporates the catalyzing effect of government subsidies. The presence of this linear relationship underscores that subsidies serve not merely as cost-reduction instruments, but as mechanisms that reshape firms' cost–benefit structures and bolster their capacity to manage carbon price risk. By providing a financial buffer, green subsidies enhance resilience against carbon price volatility and support more ambitious investments in low-carbon innovation and long-term green initiatives. For managers, this suggests that the primary objective of seeking and utilizing subsidies should shift from simple cost reduction toward strengthening strategic flexibility and mitigating carbon-related risk exposure. Companies need to integrate carbon price jumps into their strategic scenario analysis framework to enhance the resilience of their supply chains against disruptions and the flexibility of their investment structures.

Period I The government chooses the optimal green subsidy $s^a(t)$ to maximize the social welfare $W^a(t)$ in Eq. (6), which is

$$\max_{s^a(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot W^a(t) dt, \quad (18)$$

subject to Eq. (1). Define $J_W^a[\tau, c(\tau)] = \max_{s_a(t)} E_\tau \int_\tau^{+\infty} e^{-rt} \cdot W^a(t) dt$, and $J_W^a[\tau, c(\tau)] = e^{-r\tau} \cdot I_W^a[c(\tau)]$ where $I_W^a[c(\tau)] = m_1 c^2(\tau) + m_2 c(\tau) + m_3$ is the quadratic value function of the government and m_1, m_2, m_3 are non-zero parameters. The HJB equation of the government in the basic model is

$$\max_{s^a(\tau)} \left[e^{-r\tau} \cdot W^a(\tau) + \mathcal{A}^{s_a(\tau)} J_W^a[\tau, c(\tau)] \right] = 0. \quad (19)$$

Substituting Equations (C.1) and (C.2) in Appendix C.1 back to Eq. (19), the HJB equation becomes

$$\begin{aligned} \max_{s^a(\tau)} r[m_1 c^2(\tau) + m_2 c(\tau) + m_3] &= \underbrace{a_1 s^a(\tau) - a_2 [s^a(\tau)]^2}_{\text{The green utility}} \\ &- \underbrace{L[u_i q_i^a(\tau) + u_j q_j^a(\tau) - n v_i^a(\tau)]}_{\text{The social damage of carbon emissions}} \\ &+ \underbrace{[2m_1 c(\tau) + m_2][c(\tau) - \phi s^a(\tau)] + m_1 \sigma^2 c^2(\tau)}_{\text{Drift term}} + \underbrace{\int_{z \in \mathbb{R}} m_1 \gamma^2(\tau, z) c^2(\tau) \eta(dz)}_{\text{Diffusion term}} + \underbrace{\int_{z \in \mathbb{R}} m_1 \gamma^2(\tau, z) c^2(\tau) \eta(dz)}_{\text{Jump-diffusion term}}. \end{aligned} \quad (20)$$

The optimal green subsidy in the basic model is given by solving Equation (20),

$$s^a(\tau) = -\frac{m_1 \phi}{a_2} c(\tau) + \frac{a_1 - m_2 \phi}{2a_2}. \quad (21)$$

The parameters $k_1, k_2, k_3, g_1, g_2, g_3$, and m_1, m_2, m_3 , can be calculated by solving the coupled HJB Eqs. (10), (15), and (20) displayed in Appendix C.1. The optimal dynamic green subsidy $s^a(t)$ and the stochastically dynamic carbon price $C(t)$ is still obeying the optimal condition in Eq. (21). The model explicitly incorporates the jump–diffusion characteristics of carbon prices, indicating that external shocks (such as technological breakthroughs, sudden changes in climate policy, and energy crises) have become critical factors that cannot be ignored

in managerial decision-making. Governments should establish subsidy mechanisms linked to carbon price jumps to provide effective risk buffering for firms, thereby incentivizing enterprises to maintain emission reduction investments in highly uncertain environments. The optimal subsidy is no longer a fixed value or a piecewise constant but rather a continuous response function that adjusts dynamically with stochastic carbon prices. This implies that effective environmental policies must possess market responsiveness and state dependency. Governments should not formulate rigid subsidy schemes but instead establish dynamic subsidy mechanisms that are coordinated in real-time with the carbon market, thereby enhancing the agility of policy responses to market fluctuations. Carbon prices with jump components serve as a crucial nexus linking government policy and corporate strategy. Through the market-based signal of carbon pricing, governments can indirectly yet precisely guide corporate emission reduction actions, reducing distortions caused by administrative intervention. The linear functional relationship between subsidies and carbon prices essentially constitutes a public strategy rule, reducing strategic uncertainty between the government and firms. Enterprises can use this rule to predict government actions, thereby conducting long-term low-carbon investments with greater confidence. The government can enhance the credibility of its policies and improve their effectiveness by committing to such a rule. The linear function between green subsidies and carbon prices is highly computable and operable, providing the government with a clear and transparent policy tool while offering firms a stable expectation framework. This helps eliminate information asymmetry and game-theoretic misalignment between the government and enterprises, facilitating a shift toward modelable and optimizable collaborative emission reduction, thereby improving the efficiency of the overall carbon governance system.

Proposition 1. The explicit solution of the stochastic carbon price $C(t)$ in the basic model denoted as $C^a(t)$ is given as

$$\begin{aligned} C^a(t) = \exp \left(\underbrace{\left(\frac{m_1 \phi + a_2}{a_2} - \frac{1}{2} \sigma^2 \right) t}_{\text{Drift term}} + \underbrace{\sigma B(t)}_{\text{Diffusion term}} \right) \\ + \underbrace{\int_0^t \int_{z \in \mathbb{R}} \left[\ln[1 + \gamma(v, z)] - \gamma(v, z) \right] \eta(dz) dv}_{\text{The drift part of jump-diffusion term}} \\ + \underbrace{\int_0^t \int_{z \in \mathbb{R}} \ln[1 + \gamma(v, z)] \tilde{N}(dv, dz)}_{\text{Jump-diffusion term}} \cdot \left(C_0 + \underbrace{\frac{\phi(m_2 \phi - a_1)}{2a_2} \left[\left(\sigma - \frac{m_1 \phi + a_2}{a_2} \right) t}_{\text{Drift term}} \right. \right. \\ \left. \left. - \underbrace{\sigma dB(t)}_{\text{Diffusion term}} + \underbrace{\int_0^t \int_{z \in \mathbb{R}} \frac{\gamma(v, z)}{1 + \gamma(v, z)} \eta(dz) dv}_{\text{The drift part of jump-diffusion term}} - \underbrace{\int_0^t \int_{z \in \mathbb{R}} \frac{\gamma(v, z)}{1 + \gamma(v, z)} \tilde{N}(dv, dz)}_{\text{Jump-diffusion term}} \right) \right]. \end{aligned} \quad (22)$$

Proof. The stochastic differential equation of carbon price is displayed in Equation(A.2), Appendix A. $\square$

As the government plays a leadership role in Stackelberg stochastic differential game, the carbon price is determined by the optimal equilibrium feedback control of the leader. The carbon price uncertainty is impacted by green subsidy $s^a(t)$ from the government, which indicates that carbon quotas trading is a typical policy-oriented abatement instrument.

Corollary 2. The expected optimal carbon price $E[C^a(t)]$ is calculated in Appendix A, which is

$$E[C^a(t)] = O_1^a \exp[O_2^a t] + O_3^a. \quad (23)$$

Substituting Eq. (23) back to Eq. (21), we have the expected optimal green subsidy $E[s^a(t)]$,

$$E[s^a(t)] = O_4^a \exp[O_2^a t] + O_5^a. \quad (24)$$

Substituting Eq. (11) back to Eq. (23), we have the expected optimal abatement $E[v_i^a(t)]$ of the carbon quotas seller firm,

$$E[v_i^a(t)] = O_6^a \exp[O_2^a t] + O_7^a. \quad (25)$$

Substituting Eq. (17) back to Eq. (23), we have the expected optimal outputs $E[q_i^a(t)]$ and $E[q_j^a(t)]$,

$$E[q_i^a(t)] = O_8^a \exp[O_2^a t] + O_9^a, \quad E[q_j^a(t)] = O_{10}^a \exp[O_2^a t] + O_{11}^a, \quad (26)$$

where the parameters $O_1^a - O_{11}^a$ are displayed in Table D.2 in Appendix D.2.

Note that when computing the control variable's expected value, we consistently substitute the deterministic carbon price $c(t)$ in the control variable expressions in Eqs. (11), (17), and (21) with the stochastic carbon price $C(t)$ throughout the calculation of Corollary 2. The explicit solutions of optimal closed-loop control strategies of the government and the carbon trading firms discussed above provide the decision-maker with a computable approach to determine the operational objectives in the carbon-constrained scenario. The total carbon emissions are traceable with the optimal outputs and abatement inputs from the firms. The achievement of carbon reduction objectives is the essential standard to evaluate the effectiveness of carbon emission reduction policies. Proposition 1 and Corollary 2 illustrate that the optimal control functions not only outline the government's optimal subsidy strategies but also precisely calculate the optimal production levels and emissions reduction investments for enterprises. In this stochastic differential game, the government assumes the pivotal role of a leader providing crucial emission reduction subsidies to low-emission enterprises. The strategic interactions between the government and enterprises determine the optimal path for societal green transformation under policy guidance. As the government adjusts its optimal subsidy levels, enterprises must adapt their production and emissions reduction strategies accordingly. This constant interplay of carbon emission reduction policies and reactive strategies shapes the pattern of the carbon trading market.

5. Extension model of principal–agent problems

The sequence of decision-making in the green transition Stackelberg stochastic differential game with principal–agent extension is presented in Fig. 2.

- The government determines the optimal green subsidy $s^b(t)$ based on social welfare maximization goal with optional strategies {Regulated, Unregulated}.
- The buyer firm j chooses the optional strategies {Honest, Dishonest} and decides whether increase abatement input $v_j^b(t)$.
- The carbon quotas seller firm i determines the optimal abatement input $v_i^b(t)$ with given green subsidy $s^b(t)$.
- The carbon trading firms i and j make optimal outputs $q_i^b(t), q_j^b(t)$ with profit maximization objectives.
- The price $p^b(t)$ of products is determined by outputs.
- The firm i sells surplus carbon quotas $[A - D_i^b(t)]$, while the firm j honestly buys carbon emissions $[D_j^b(t) - A]$ or disguises as the firm i selling $[A - D_j^b(t)]$.

5.1. Second decision-making period of the carbon seller

Period II The maximization objectives of all game players are subject to the renewed dynamic carbon price, which is

$$\begin{cases} dC(t) = \underbrace{[C(t) - \phi s^b(t)]dt}_{\text{Drift term}} + \underbrace{\sigma C(t)dB(t)}_{\text{Diffusion term}} + \underbrace{C(t) \cdot \int_{z \in \mathbb{R}} \gamma(t, z) \tilde{N}(dt, dz)}_{\text{Jump-diffusion term}}, \\ C(0) = C_0, \end{cases} \quad (27)$$

where $s^b(t)$ is the green subsidy in the principal–agent condition. The price of products becomes $p^b(t) = \epsilon - Q^b(t)$ where the total output is $Q^b(t) = q_i^b(t) + q_j^b(t)$. For the seller firm i , the optimizing objective is the profit function $\Pi_i^b(t)$ in the infinite horizon by choosing the optimal the control variables $q_i^b(t), v_i^b(t)$ in the extension model, which is

$$\max_{q_i^b(t), v_i^b(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot \Pi_i^b(t) dt, \quad (28)$$

subject to Eq. (27). Define $J_i^b[\tau, c(\tau)] = \max_{q_i^b(\tau), v_i^b(\tau)} E_\tau \int_\tau^{+\infty} e^{-r\tau} \cdot \Pi_i^b(t) dt$, and $J_i^b[\tau, c(\tau)] = e^{-r\tau} \cdot I_i^b[c(\tau)]$ where the quadratic value function is $I_i^b[c(\tau)] = K_1 c^2(\tau) + K_2 c(\tau) + K_3$. K_1, K_2, K_3 are non-zero parameters of the value function $I_i^b[c(t)]$. The scripts in the value function $I_i^b[c(t)]$ represent the profit in the extended model (the superscript I^b) of firm i . The HJB equation of the carbon seller is

$$\max_{q_i^b(\tau), v_i^b(\tau)} \left[e^{-r\tau} \cdot \Pi_i^b(\tau) + \mathcal{A}^{q_i^b(\tau), v_i^b(\tau)} J_i^b[\tau, c(\tau)] \right] = 0, \quad (29)$$

where the operator $\mathcal{A}^{q_i^b(\tau), v_i^b(\tau)} J_i^b[\tau, c(\tau)]$ is in Equation (C.40), Appendix C.2. Applying the dynamic programming principle, the HJB equation of firm i in the extended model becomes

$$\begin{aligned} & \max_{q_i^b(\tau), v_i^b(\tau)} r[K_1 c^2(\tau) + K_2 c(\tau) + K_3] \\ &= \underbrace{[\epsilon - q_i^b(\tau) - q_j^b(\tau)]q_i^b(\tau) + c(\tau) \cdot [A + n_i v_i^b(\tau) - u_i q_i^b(\tau)]}_{\text{Production profits}} \\ & \quad - \underbrace{b_i [v_i^b(\tau)]^2 - h_i [q_i^b(\tau)]^2}_{\text{Costs}} + \underbrace{s^b(\tau)}_{\text{Green subsidy}} + \underbrace{[2K_1 c(\tau) + K_2][c\tau - \phi s^b(\tau)]}_{\text{Drift term}} \\ & \quad + \underbrace{K_1 \sigma^2 c^2(\tau)}_{\text{Diffusion term}} + \underbrace{\int_{z \in \mathbb{R}} K_1 \gamma^2(\tau, z) c^2(\tau) \eta(dz)}_{\text{Jump-diffusion term}} \end{aligned} \quad (30)$$

We obtain the optimal abatement input $v_i^b(\tau)$ and the linear equation of the optimal outputs $q_i^b(\tau), q_j^b(\tau)$ of firm i by solving Equation (30), which are

$$v_i^b(\tau) = \frac{n_i}{2b_i} c(\tau). \quad (31)$$

$$q_i^b(\tau) = -\frac{u_i}{2(h_i + 1)} c(\tau) + \frac{\epsilon - q_j^b(\tau)}{2(h_i + 1)}. \quad (32)$$

5.2. Principal–agent issues with asymmetric information

This section extends the stochastic dynamic game in the principal–agent circumstance where the carbon quotas buyer displays complex moral hazard behaviors upon carbon abatement. The government is the principal, and carbon trading firms are agents. There are two independent events X, Y . X means the firm j implements abatement input honestly or not, while Y represents whether the government regulates high-carbon firms or not. $X = 0$ stands for the firm exactly reducing carbon emissions with constant probability x where $0 \leq x \leq 1$, while $X = 1$ means the firm disguises in carbon reduction with probability $(1 - x)$.

$$X = \begin{cases} 0 & \text{with probability } x, \\ 1 & \text{with probability } 1 - x. \end{cases} \quad (33)$$

$Y = 0$ represents the government regulate firms with constant probability y where $0 \leq y \leq 1$, while $Y = 1$ means that the government provides green subsidy without regulations with probability $(1 - y)$.

$$Y = \begin{cases} 0 & \text{with probability } y, \\ 1 & \text{with probability } 1 - y. \end{cases} \quad (34)$$

The carbon trading firms are free of charge on carbon emissions within the initial carbon quota of A . The low-carbon firm can gain carbon

Table 1
The payoff matrix.

The game	Carbon quotas buyer firm	
	Honest $P(X = 0) = x$	Dishonest $P(X = 1) = 1 - x$
Government		
Regulated $P(Y = 0) = y$	$[w_{11}^b(t), \pi_{11}^b(t)]$	$[w_{12}^b(t), \pi_{12}^b(t)]$
Unregulated $P(Y = 1) = 1 - y$	$[w_{21}^b(t), \pi_{21}^b(t)]$	$[w_{22}^b(t), \pi_{22}^b(t)]$

trading profits by selling extra carbon quota $C(t)[D_i^b(t) - A]$, while the high-carbon firm should buy carbon quotas with the carbon trading cost $C(t)[A - D_j^b(t)]$. The principal accelerates the green transition with social welfare maximization and carbon neutrality objectives by providing green subsidies to encourage firms to reduce carbon emissions. The agents make production and abatement decisions with profit-maximizing objectives incentivized by green subsidies. The core challenge is to explore an effective incentive compatibility mechanism to unify the goals of the government and firms. The carbon quotas buyer intends to disguise himself as a low-carbon enterprise that can sell carbon quotas to gain green subsidies from the government. The moral hazard of the carbon trading buyers leads to a renewed non-cooperative Stackelberg stochastic differential game that the government should choose the optimal green subsidy $s^b(t)$, while the firms accordingly decide the optimal productions $q_j^b(t)$, $q_j^b(t)$ and abatement inputs $v_i^b(t)$, $v_j^b(t)$. The carbon trading profit $C(t)[D_i^b(t) - A]$, carbon quotas buying cost $C(t)[A - D_j^b(t)]$, green utility $U[s^b(t)]$, and the damage function $L[D_i^b(t) + D_j^b(t)]$ are publicly available common knowledge in effective environmental information disclosure system. Assuming that whether the buyer pretends to be a low-carbon firm and whether the government regulates the carbon trading market is private information.

The strategic sets for carbon quotas buyer firm and the government are $\Omega_x = \{\text{Honest}, \text{Dishonest}\}$, $\Omega_y = \{\text{Regulated}, \text{Unregulated}\}$, respectively. If the buyer firm chooses to reduce carbon emissions honestly, it means that the high-carbon firm will invest in green technological innovations and increase renewable energy consumption. The abatement input $v_j^b(t)$ of the buyer firm j is the additional term in the profit-maximizing function. If the buyer firm chooses to reduce carbon emissions dishonestly and defraud the government gain the green subsidy, which means that the firm maintains its original manufacturing type without abatement input. The real carbon emission functions of the honest and dishonest buyer states are $D_j^{b1}(t) = u_j q_j^b(t) - n_j v_j^b(t)$ and $D_j^{b2}(t) = u_j q_j^b(t)$, respectively. If the government chooses to regulate the carbon quotas buyers with constant regulation cost R , which means that the green subsidy $s^b(t)$ will be distributed in an impartial method according to whether the firms increase abatement inputs. Environmental regulations represent that the government will estimate the total carbon emissions of the buyer firms and establish carbon disclosure reports. If the firm is investigated as an honest one with regulatory certification from the government, the firm can gain green subsidy $s^b(t)$ from the government and green premium H from consumers' low-carbon preference, while the dishonest firm has to pay a penalty F in dishonest abatement behaviors without green subsidy. R, H, F are deterministic constants. If the government chooses to provide green subsidies without environmental regulation, the carbon quotas buyer and seller can both gain subsidies equally. The payoff matrix of the government and carbon quotas buyer are displayed in Table 1.

There are four strategic elements in the payoff matrix displayed in Table 2 to capture interactions between the regulator and producers. Firstly, (Honest, Regulated) strategy with probability xy and a government–enterprise payoff group $[w_{11}^b(t), \pi_{11}^b(t)]$, where the buyer increases abatement input and the government provides green subsidy with regulation cost. The payoff function of the buyer firm is $[p^b(t) + H]q_j^b(t) - h_j[q_j^b(t)]^2 - b_j[v_j^b(t)]^2 + s^b(t) + C(t)[A - D_j^{b1}(t)]$ with positive green premium in products price $p^b(t) = \epsilon - q_i^b(t) - q_j^b(t)$. The additional abatement cost of firm j is $b_j[v_j^b(t)]^2$, where $v_j^b(t)$ is the abatement input and b_j is the positive abatement efficiency parameter. The payoff

Table 2
The list of notations and payoffs.

Notations	Payoffs
$\pi_{11}^b(t)$	$[p^b(t) + H]q_j^b(t) - h_j[q_j^b(t)]^2 - b_j[v_j^b(t)]^2 + s^b(t) + C(t)[A - D_j^{b1}(t)]$
$w_{11}^b(t)$	$U[2s^b(t)] - L[D_i^b(t) + D_j^{b1}(t)] - R$
$\pi_{12}^b(t)$	$p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 - F - C(t)[D_j^{b2}(t) - A]$
$w_{12}^b(t)$	$\theta U[s^b(t)] - \lambda L[D_i^b(t) + D_j^{b2}(t)] - R + F$
$\pi_{21}^b(t)$	$p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 - b_j[v_j^b(t)]^2 + s^b(t) + C(t)[A - D_j^{b1}(t)]$
$w_{21}^b(t)$	$U[2s^b(t)] - L[D_i^b(t) + D_j^{b1}(t)]$
$\pi_{22}^b(t)$	$p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 + s^b(t) - C(t)[D_j^{b2}(t) - A]$
$w_{22}^b(t)$	$\theta U[2s^b(t)] - \lambda L[D_i^b(t) + D_j^{b2}(t)]$

function of the government is $U[2s^b(t)] - L[D_i^b(t) + D_j^{b1}(t)] - R$, where the green utility becomes $U[2s^b(t)]$ with double subsidies, where $U[s^b(t)] = a_1 s^b(t) - a_2 [s^b(t)]^2$. The green utility describes the environmentally friendly benefits, such as the delay of increasing temperature, optimizing the energy-consuming structure, and alleviating the greenhouse effect. Social welfare is increased from firms' honest carbon abatement endeavors. Secondly, (Dishonest, Regulated) with probability $(1 - x)y$ and a government–enterprise payoff group $[w_{12}^b(t), \pi_{12}^b(t)]$, where the buyer pretends to invest in carbon reduction and the government investigates the buyer firm states in reality with positive regulation cost. The payoff function of the buyer firm is $p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 - F - C(t)[D_j^{b2}(t) - A]$, where the buyer firm loses the green subsidy and pays the additional penalty. The payoff function of the government is $\theta U[s^b(t)] - \lambda L[D_i^b(t) + D_j^{b2}(t)] - R + F$, where the discounted green utility is $\theta U[s^b(t)]$ with a parameter $0 < \theta < 1$ and the augmentative damage function becomes $\lambda L[D_i^b(t) + D_j^{b2}(t)]$ with an extended social cost of carbon ratio $\lambda > 1$. The total social welfare decreases owing to a lower green utility and a higher environmental damage.

Thirdly, (Honest, Unregulated) with the constant probability $x(1 - y)$ and a government–enterprise payoff group $[w_{21}^b(t), \pi_{21}^b(t)]$, where the buyer firm invests carbon abatement with the profit function $p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 - b_j[v_j^b(t)]^2 + s^b(t) + C(t)[A - D_j^{b1}(t)]$ and the government provides green subsidy without regulation with payoff function $U[2s^b(t)] - L[D_i^b(t) + D_j^{b1}(t)]$. Finally, (Dishonest, Unregulated) with probability $(1 - x)(1 - y)$ and a government–enterprise payoff group $[w_{22}^b(t), \pi_{22}^b(t)]$, where the buyer firm chooses to disguise with a profit $p^b(t)q_j^b(t) - h_j[q_j^b(t)]^2 + s^b(t) - C(t)[D_j^{b2}(t) - A]$, where the firm j pretend successfully as a low-carbon firm gaining green subsidy and selling carbon quotas profit $C(t)[A - D_j^{b2}(t)]$. The government does not regulate companies' behaviors with the payoff function $\theta U[2s^b(t)] - \lambda L[D_i^b(t) + D_j^{b2}(t)]$. The double green subsidies without environmental regulation lead to a discounted green utility $\theta U[2s^b(t)]$ and an augmentative social damage of carbon emissions $\lambda L[D_i^b(t) + D_j^{b2}(t)]$, while the buyer firm cannot gain the green premium derived from the low-carbon authentication.

5.3. Stochastic optimization on differential game with principal–agent extension

Considering the payoff matrix in Table 1, the profit function of the firm j is

$$\begin{aligned} \bar{\Pi}_j^b(t) &= \pi_{11}^b(t) \cdot P[X = 0, Y = 0] + \pi_{12}^b(t) \cdot P[X = 1, Y = 0] \\ &\quad + \pi_{21}^b(t) \cdot P[X = 0, Y = 1] + \pi_{22}^b(t) \cdot P[X = 1, Y = 1]. \end{aligned} \quad (35)$$

For X, Y are independent events, there is $P[X = 0, Y = 0] = P(X = 0) \cdot P(Y = 0) = xy$. And the profit function becomes $\bar{\Pi}_j^b(t) = xy\pi_{11}^b(t) + (1-x)y\pi_{12}^b(t) + x(1-y)\pi_{21}^b(t) + (1-x)(1-y)\pi_{22}^b(t)$. The optimization objective of carbon quotas buyer becomes

$$\max_{q_j^b(t), v_j^b(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot \bar{\Pi}_j^b(t) dt, \quad (36)$$

subject to Eq. (27), where $\bar{\Pi}_j^b(t)$ is displayed in Appendix C.2, Equation (C.26). Eq. (36) indicates that the profit of firm j in the principal-agent scenario becomes more complicated with conditional probabilities related to the behavioral uncertainties of the government and the high-carbon firms. Define $J_j^b[\tau, c(\tau)] = \max_{q_j^b(t), v_j^b(t)} E_\tau \int_\tau^{+\infty} e^{-rt} \cdot \bar{\Pi}_j^b(t) dt$, and $J_j^b[\tau, c(\tau)] = e^{-r\tau} \cdot I_j^b[c(\tau)]$ where $I_j^b[c(\tau)] = G_1 c^2(\tau) + G_2 c(\tau) + G_3$. G_1, G_2, G_3 are non-zero parameters of the value function $I_j^b[c(\tau)]$. The HJB equation of the firm j in the extension model is

$$\max_{q_j^b(\tau), v_j^b(\tau)} \left[e^{-r\tau} \cdot \bar{\Pi}_j^b(\tau) + \mathcal{A}^{q_j^b(\tau)} J_j^b[\tau, c(\tau)] \right] = 0. \quad (37)$$

The behavioral uncertainty of the carbon quotas buyer increases the game space where the government and the firm j interactions have priority over the carbon trading of two firms.

We obtain the optimal control of abatement input and the equation of the optimal output in the extension model for firm j , which are

$$v_j^b(\tau) = \frac{n_j}{2b_j} c(\tau), \quad (38)$$

$$q_j^b(\tau) = -\frac{u_j}{2(h_j + 1)} c(\tau) + \frac{Hxy + \epsilon - q_i^b(\tau)}{2(h_j + 1)}. \quad (39)$$

Corollary 3. Substituting Eq. (39) back to Eq. (32), the optimal productions are

$$\begin{cases} q_i^b(\tau) = -\frac{(2h_j u_i + 2u_i - u_j)}{4h_i h_j + 4h_i + 4h_j + 3} c(\tau) - \frac{Hxy - 2\epsilon h_j - \epsilon}{4h_i h_j + 4h_i + 4h_j + 3}, \\ q_j^b(\tau) = \frac{(-2h_i u_j + u_i - 2u_j)}{4h_i h_j + 4h_i + 4h_j + 3} c(\tau) + \frac{2Hhixy + 2Hxy + 2\epsilon hi + \epsilon}{4h_i h_j + 4h_i + 4h_j + 3}. \end{cases} \quad (40)$$

The optimal outputs of the carbon trading firms are linear functions of carbon price $c(\tau)$ and depend on the additional constant parameters like green premium H , the probability of honest abatement x , and the probability of implementing regulation y . We can obtain the optimal feedback control function considering the camouflage strategies employed by carbon trading firms. Theoretical derivations indicate that these camouflage strategies do not alter the linear relationship between the optimal control function and the carbon price. The optimal output of carbon trading enterprises incorporates the constant probability of high-emission firms choosing disguise strategies and the government opting for a regulatory approach, representing the outcome of decision-makers' rational trade-offs under the constraint of carbon prices with jumps. For corporate managers, the optimal strategy necessitates the establishment of a dynamic risk assessment framework, treating disguise behavior as a strategic risk requiring proactive management. It also demands continuous calibration of optimal decisions based on fluctuations in parameters such as carbon prices and green premiums, thereby transforming static strategies into a dynamically adjustable optimal strategy. For government regulators, this implies that corporate behavior is predictable and can be guided through policy instruments to standardize carbon market behavior. By indirectly influencing the green premium through emission reduction policies, regulation can shift from ex-post penalties for excessive emissions to a forward-looking mechanism design. Furthermore, the robust linear relationship between optimal output and carbon prices indicates that clear decision-making rules persist even amid strategic interactions between carbon trading enterprises and the government.

Period I In the extended model, the profit function of the government is

$$\bar{W}^b(t) = w_{11}^b(t) \cdot P[X = 0, Y = 0] + w_{12}^b(t) \cdot P[X = 1, Y = 0] + w_{21}^b(t) \cdot P[X = 0, Y = 1] + w_{22}^b(t) \cdot P[X = 1, Y = 1]. \quad (41)$$

The optimization objective becomes

$$\max_{s^b(t)} E_0 \int_0^{+\infty} e^{-rt} \cdot \bar{W}^b(t) dt. \quad (42)$$

subject to Eq. (27), where the optimizing objective function of the government $\bar{W}^b(t)$ in the extension model is displayed in Appendix C.2, Equation (C.28). On the condition of the strategic choices of the high-carbon firm and the government, the stochastic optimization becomes more complicated compared with the optimizing objectives in the basic model in Section 4. Define $J_W^b[\tau, c(\tau)] = \max_{s^b(t)} E_\tau \int_\tau^{+\infty} e^{-rt} \cdot \bar{W}^b(t) dt$, and $J_W^b[\tau, c(\tau)] = e^{-r\tau} \cdot I_W^b[c(\tau)]$ where $I_W^b[c(\tau)] = M_1 c^2(\tau) + M_2 c(\tau) + M_3$. M_1, M_2, M_3 are non-zero parameters of the value function $I_W^b[c(\tau)]$. The HJB equation of the government with principal-agent extension is

$$\max_{s^b(\tau)} \left[e^{-r\tau} \cdot \bar{W}^b(\tau) + \mathcal{A}^{s^b(\tau)} J_W^b[\tau, c(\tau)] \right] = 0. \quad (43)$$

The parameters $M_1, M_2, M_3, K_1, K_2, K_3$, and G_1, G_2, G_3 are calculated in Appendix C.2.

Corollary 4. We obtain the optimal green subsidy by solving Equation (43) in the extended model,

$$s^b(\tau) = -\frac{\phi M_1}{a_2[\theta(3y-4)(x-1)+4x]} c(\tau) + \frac{a_1(x+1) - \phi M_2}{2a_2[\theta(3y-4)(x-1)+4x]}. \quad (44)$$

Eqs. (40) and (44) remain valid in the extended model, which incorporates dynamic production levels $q_i^b(t)$ and $q_j^b(t)$, the green subsidy $s^b(t)$, and the stochastic carbon price $C(t)$. By introducing a probability matrix that captures firms' disguise strategies and government regulatory strategies, our model preserves an explicit linear relationship among the optimal output, dynamic green subsidies, and the jumping carbon price, even when accounting for complex strategic interactions. This result demonstrates that the model retains both mathematical tractability and economic interpretability under more realistic assumptions including asymmetric information and strategic behaviors, thereby significantly enhancing the structured characterization of real-world carbon market mechanisms. From a managerial perspective, the model highlights the potential for finely tuned policy regulation even in highly information-asymmetric market environments. The government can formulate a composite strategy that integrates dynamic subsidies with stochastic regulation, establishing an adaptive governance mechanism driven by real-time information updates. This mechanism dynamically adjusts intervention intensity based on carbon price fluctuations and firms' behavioral probabilities. It leverages credible regulatory threats to effectively curb short-term disguise incentives, thereby steering firms toward prioritizing long-term emission reduction outcomes and reputation capital accumulation.

Furthermore, the computable feedback control strategies provide clear operational guidelines for bidirectional interactions between government and enterprises. The government can optimize subsidy levels and monitoring frequency in real time based on changes in market state variables, establishing a closed-loop control framework for policy intervention. Firms, under clearer policy expectations, can more accurately plan emission reduction investments and technology transition roadmaps, avoiding investment delays or resource misallocation caused by policy uncertainty. Although behavioral uncertainties are incorporated, the stability of the optimal linear relationship significantly enhances policy predictability and coordination efficacy. This facilitates more transparent public policy through rule-based and formulaic

Table 3
Parameters and values.

Parameters	β	κ	α_1	α_2	λ	σ	ϕ	C_0	r	A
Values	0.2	3	10	5	1.2	0.005	0.16	56	0.02	60
Parameters	n_i	n_j	b_i	b_j	h_i	h_j	u_i	u_j	a_1	a_2
Values	0.85	0.65	1.2	1.5	0.96	0.95	0.06	0.08	2.5	2.2
Parameters	F	H	θ	x	y	γ	L	R	ϵ	t
Values	1000,000	100	0.5	0.8	0.6	1	150,000	10,000	500	[0,5]

subsidy mechanisms, while also encouraging firms to focus on technological innovation and operational efficiency rather than strategic gaming. The model provides a theoretical foundation and methodological support for constructing an incentive-compatible mechanism for collaborative emission reduction between the government and firms. It demonstrates that well-designed policy mixes can effectively reconcile interest conflicts in asymmetric information environments, encouraging firms to establish a positive feedback loop of continuous emission reduction, obtaining policy incentives, and reinvesting in green technologies. Thereby, it improves the overall efficiency of the carbon market and promotes a transition in environmental governance from passive compliance to active participation and collaborative innovation toward achieving carbon neutrality goals.

Proposition 2. We obtain the explicit solution of the stochastic carbon price $C(t)$ in the extension model denoted as $C^b(t)$, which is

$$\begin{aligned}
 C^b(t) = & \exp \left[\underbrace{\left(\frac{\phi^2 M_1}{a_2[\theta(3y-4)(x-1)+4x]} + 1 - \frac{1}{2}\sigma \right)t}_{\text{Drift term}} + \underbrace{\sigma B(t)}_{\text{Diffusion term}} \right] \\
 & + \underbrace{\int_0^t \int_{\mathbb{R}} \left[\ln[1 + \gamma(v, z)] - \gamma(v, z) \right] \eta(dz) dv}_{\text{The drift part of jump-diffusion term}} + \underbrace{\int_0^t \int_{\mathbb{R}} \ln[1 + \gamma(v, z)] \tilde{N}(dv, dz)}_{\text{Jump-diffusion term}} \\
 & \cdot \left[C_0 + \underbrace{\frac{\phi(a_1\theta x + M_2\phi - a_1\theta - 2a_1x)}{2a_2(3\theta xy - 4\theta x - 3\theta y + 4\theta + 4x)}}_{\text{Drift term}} \left(\left[\sigma - \frac{\phi^2 M_1}{a_2[\theta(3y-4)(x-1)+4x]} \right] t \right) \right. \\
 & \left. + \underbrace{\frac{-\sigma B(t)}{2}}_{\text{Diffusion term}} + \underbrace{\int_0^t \int_{\mathbb{R}} \frac{\gamma(v, z)}{1 + \gamma(v, z)} \eta(dz) dv}_{\text{The drift part of jump-diffusion term}} - \underbrace{\int_0^t \int_{\mathbb{R}} \frac{\gamma(v, z)}{1 + \gamma(v, z)} \tilde{N}(dv, dz)}_{\text{Jump-diffusion term}} \right]. \quad (45)
 \end{aligned}$$

Proof. Substituting Eq. (44) back to Eq. (27), we obtain the stochastic differential equation of carbon price with optimized green subsidy with principal-agent extension displayed in Equation (B.1), Appendix B. $\square$

Compared with Eq. (22), the explicit solution of the carbon price in the extension model illustrates that the pretending and regulating probabilities x, y determine the optimal green subsidy from the government in Eq. (44), and then exert a significant influence on the carbon price $C^b(t)$ in the principal-agent scenario. The advanced modeling approach considering the behavioral uncertainties of the game players in the carbon trading market improves the precision of quantified evaluating the dynamic carbon price.

Corollary 5. The expected optimal carbon price in the extension model is

$$E[C^b(t)] = O_1^b \exp[O_2^b t] + O_3^b. \quad (46)$$

Substituting Eq. (46) to Eq. (44), we obtain the expected optimal green

$$\text{subsidy } E[s^b(t)] = O_4^b \exp[O_2^b t] + O_5^b. \quad (47)$$

The expected optimal outputs $E[q_i^b(t)]$ and $E[q_j^b(t)]$ in the extension model are

$$E[q_i^b(t)] = O_6^b \exp[O_2^b t] + O_7^b, \quad E[q_j^b(t)] = W_8^b \exp[O_2^b t] + O_9^b. \quad (48)$$

The expected optimal abatement inputs $E[v_i^b(t)]$ and $E[v_j^b(t)]$ in the extension model are

$$E[v_i^b(t)] = O_{10}^b \exp[O_2^b t] + O_{11}^b, \quad E[v_j^b(t)] = O_{12}^b \exp[O_2^b t] + O_{13}^b, \quad (49)$$

where the parameters $O_1^b - O_{13}^b$ are displayed in Table D.2 in Appendix D.2.

Corollary 5 indicates that with the explicit solution of the carbon price displayed in Eq. (45), we can obtain all optimal closed-loop control strategies of the government and firms in the Stackelberg stochastic differential game with principal-agent extension. In the extended model, we analyze the underlying dynamics of game behaviors with asymmetric information. Specifically, we consider the complex behavior of enterprises that might embellish their emission reduction efforts to obtain government subsidies fraudulently. Proposition 2 and Corollary 5 illustrate that in the disguising scenario, the government faces the challenging task of authenticating genuine low-emission enterprises and distinguishing them from those who merely seek to capitalize on subsidies through deceptive means. This necessitates rigorous oversight and monitoring of the actual emission reduction achieved by enterprises, thereby significantly heightening the complexity of deriving the stochastic optimal control function. The expectation operator effectively integrates the dynamic carbon pricing mechanism's jump-diffusion parameters to formulate the optimal closed-loop control function. This sophisticated integration ensures that the optimal decisions crafted by both enterprises and the governing body explicitly account for the inherent jump risks associated with carbon pricing. As a result, this approach significantly elevates the precision and accuracy of production planning and policy formulation, contributing to more effective decision-making processes.

6. Numerical simulation

Comparative analysis of green transition strategies of carbon trading firms and the government are discussed based on the aforementioned theoretical framework in Sections 5 and 6 with numerical simulation results by programming in Matlab. The period in the stochastic differential game is $t \in [0, 5]$. The jump intensity parameter is $\kappa = 3$ which is consistent with Kou and Wang (2004). The initial carbon price is $C_0 = 56$ CNY/ton (Fan et al., 2018) with initial carbon quotas $A = 60$ ton. This paper utilizes the daily data of China's carbon market from 2022 to 2024 from the Wind database to estimate the volatility parameter and the jump intensity parameter using the GARCH method. The volatility parameter is $\sigma = 0.005$, the jump intensity parameter is $\kappa = 3$ consistent with Kou and Wang (2004), and the influential coefficient of green subsidy in carbon price is $\phi = 0.16$. We set the positive jump probability $\beta = 0.2$ with coefficients of double exponential jump-diffusion $\alpha_1 = 10, \alpha_2 = 5$. The discounting rate is $r = 0.02$. The penalty of the carbon quotas buyer dishonest abatement is $F = 1,000,000$ CNY with regulation cost $R = 100,000$ CNY for the government. The price parameter $\epsilon = 500$ and the quadratic green utility parameters are $a_1 = 2.5, a_2 = 2.2$. The green premium on honest abatement efforts of the firms is $H = 100$ CNY. The probability of the carbon buyer firm that chooses to honestly increase abatement input $v_j^b(t)$ is $x = 0.8$, while the probability of the government that chooses to regulate is $y = 0.6$. The extra social loss parameter is $\lambda = 1.2$ and the

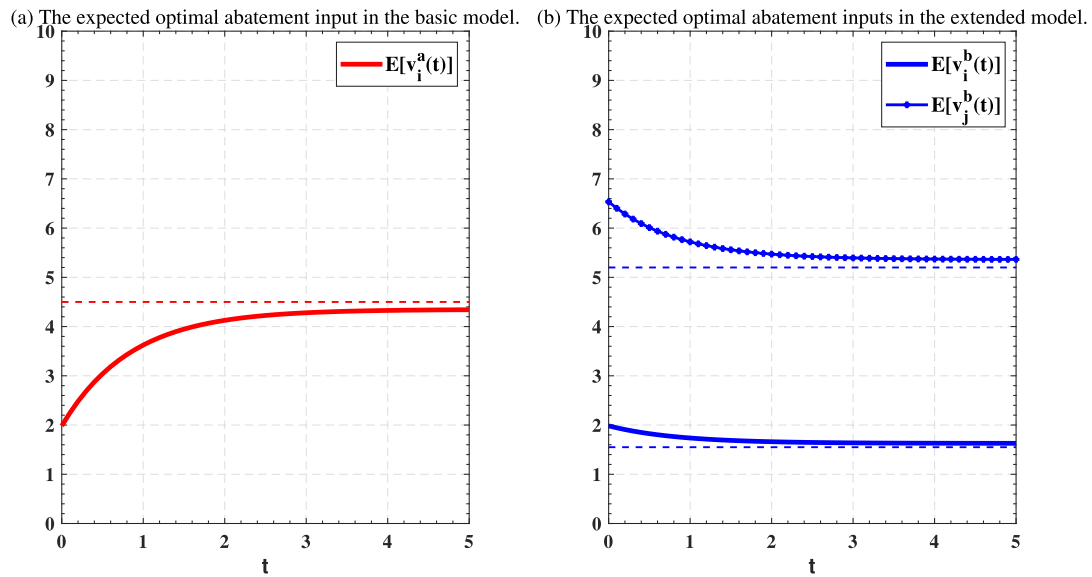


Fig. 3. The curves of the optimal abatement inputs. $E[v_i^a(t)]$ and $E[v_j^b(t)]$, $E[v_i^b(t)]$ are the optimal expected abatement inputs of the firm i in the basic and extended model. The red and blue dotted lines are stable lines for the basic and extended models.

discounted green utility ratio is $\theta = 0.5$. The values of parameters in the model are displayed in Table 3.

First, we discuss the optimal outputs and abatement inputs of carbon trading firms. Fig. 3 illustrates that in the basic model, abatement inputs of firms increase continuously in the initial period and maintain a stable level afterward. In the extended model, the abatement inputs decrease slightly and then remain stable. The extended model sheds light on a more underlying aspect of the carbon trading market where the strategic behaviors of carbon-buying (high-emission) firms that intend to disguise strategies to obtain government subsidies fraudulently displayed in Fig. 3(b). Firm j that purchases carbon quotas need to increase carbon emissions reduction inputs significantly to engage in deceptive behavior, which indirectly leads to a reduction in the actual emissions reduction inputs of firm i that sells carbon quotas. The model quantitatively assesses the cost of corporate deception. It emphasizes that the firm attempting to fraudulently obtain subsidies needs to make greater emissions reduction inputs with three times the abatement input paid by carbon sellers.

The carbon emission selling firm i has a higher production output than the carbon emission purchasing firm j , while firm j exhibits a more pronounced decreasing trend than firm i . Fig. 4 indicates that the optimal outputs fluctuate in basic and extended models. Initially, the production outputs of firms undergo a significant decrease and subsequently stabilize at a relatively constant level when neglecting deceptive behaviors. This stabilization can be attributed to their prudent approach to navigating market demand shifts and policy adjustments. Conversely, firms in the extended model with disguising strategies plateau at an elevated level. Comparing Fig. 4(a) with (b), the strategic disguise employed by carbon-purchasing enterprises has triggered a pivotal transformation in optimal production levels, shifting from being lower than those of carbon sellers to exceeding them. In contrast to Fig. 4(a), the production outputs of both the carbon emission selling firm i and the carbon emission purchasing firm j in Fig. 4(b) exhibit different trends. Their outputs initially increase and then stabilize at a relatively constant level. Carbon quota-selling enterprises must develop optimal production strategies in decision-making following the numerically simulated curve presented in Fig. 4. When disregarding carbon buyers' deceptive strategies, output progressively decreases from elevated levels compared to the carbon buyer toward 87 units. Conversely, when accounting for carbon buyers' strategic behavior, output systematically increases from a lower baseline compared to the carbon

buyer toward 93 units. These optimal output curves satisfy carbon trading equilibrium conditions, enabling both parties to achieve profit-maximizing output levels with carbon price jumps. When the carbon buyer engages in strategic misrepresentation by posing as low-emission entities, the carbon quota seller should adopt a more aggressive production strategy compared to scenarios where such deceptive practices are absent. This elevated output level serves as a countermeasure to ensure the optimal profit realization under asymmetric information conditions. In a stochastic differential game framework, the optimal strategy for enterprises transcends blind capacity expansion or short-term gain pursuit. It necessitates a dynamic adjustment of capacity investments and abatement strategies, grounded in theoretical models that enable precise responses to market signals and efficient resource allocation.

Second, we explore firms' actual carbon emissions. Fig. 5 reveals that the carbon emissions in the basic and extended scenarios have contrary tendencies. It is crucial to recognize that carbon sellers maintain a unique position of advantage due to their ability to consistently emit below their allocated quotas. Conversely, carbon buyers, often representing enterprises with higher emission profiles, find themselves exceeding their initial quotas and thus in need of purchasing additional carbon quotas. Whether they disguise their actual carbon emission status or not, the strategies of buyers ultimately fail to dislodge the carbon sellers from their entrenched position as the suppliers of carbon quotas. This stability in the market structure underscores the resilience of the carbon trading system and its ability to withstand such tactical game players. The position of carbon sellers does not fundamentally alter in the dynamic carbon trading market. Enterprises are encouraged to purchase carbon quotas based on emissions, fostering a more accurate and efficient allocation of resources within the carbon trading market.

Furthermore, the basic model demonstrates a significant initial reduction in actual carbon emissions displayed in Fig. 5(a). In contrast, the extended model displayed in Fig. 5(b) exhibits a slight increase in carbon emissions and then maintains stationary. In Fig. 5(a), the sum of carbon emissions in the basic model is greater than $2A$ in the period $t \in [0, 1]$, while they are lower than $2A$ in the period $t \in [1, 5]$. The sum of the carbon emissions in Fig. 5(b) is greater than that in Fig. 5(a) and is greater than $2A$. Fig. 5 indicates that the deceptive strategies adopted by high-emission enterprises will lead to an increase in the total amount of carbon emissions. By accounting for the intricate strategic interactions between governments and enterprises, our

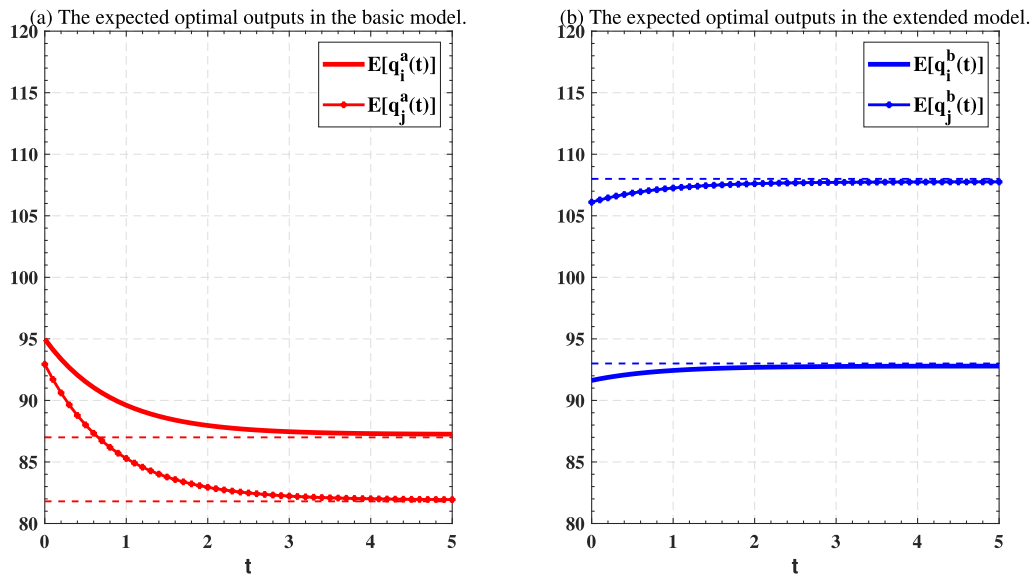


Fig. 4. The curves of the optimal expected outputs. $E[q_i^a(t)]$, $E[q_j^a(t)]$ and $E[q_i^b(t)]$, $E[q_j^b(t)]$ are the optimal expected outputs of firms i and j in the basic and extended model. The red and blue dotted lines are stable lines for the basic and extended models.

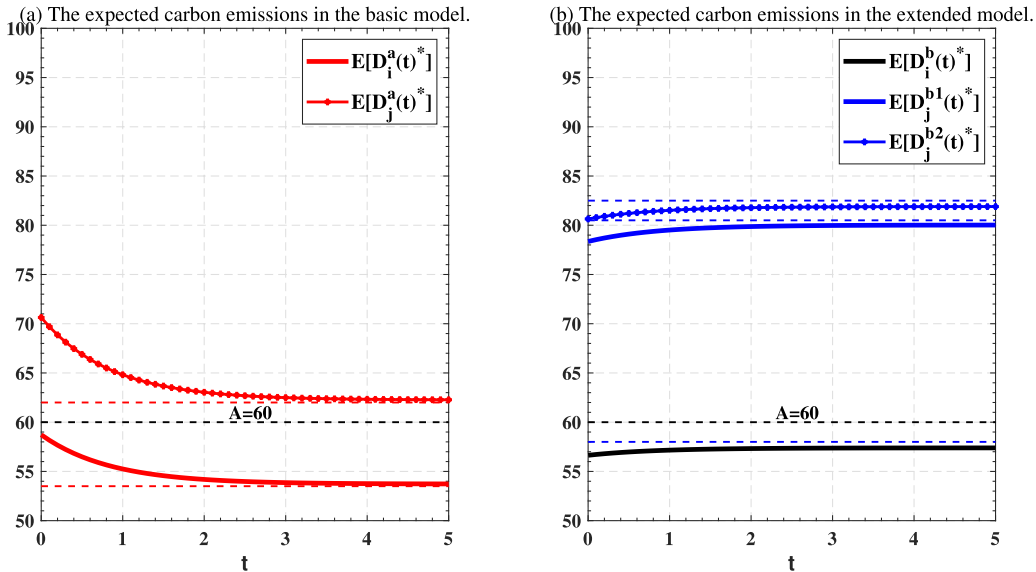


Fig. 5. The expected optimal carbon emissions of two carbon trading firms. The expected equilibrium carbon emission is $E[D_i^b(t)^*]$ of firm i and the honest and dishonest carbon emissions are denoted as $E[D_j^{b1}(t)^*]$, $E[D_j^{b2}(t)^*]$ of firm j . The red and blue dotted lines are stable lines for the basic and extended models. The black dotted lines are the contour lines for the initial carbon quotas $A = 60$.

findings reveal that even optimal decisions made by carbon-trading firms can lead to distinct trends in carbon emissions. The achievement of the emission reduction target highlights the robustness and validity of the jump–diffusion stochastic differential game-theoretic model in the policy-driven carbon trading markets. The stabilization of carbon emission levels indicates that the government's green subsidies are an effective measure for carbon emission reduction.

The supplementary numerical simulations in Appendix E uncover several key findings. The dynamic allocation of subsidies implemented by the government significantly enhances social welfare when disguised behaviors are not considered. The carbon seller achieves a higher profit in the basic model, while the carbon buyer attains greater profitability in the extended model. The optimal green subsidy as a pivotal component of the government's control strategy displays a monotonically increasing trend. Within the extended model framework that captures firms' strategic behaviors, social welfare is found to decrease as the

initial carbon quotas increase. Following an initial period of increase, the equilibrium carbon price stabilizes at a higher level. Detailed analyses of the variations in the optimal green subsidy and the equilibrium carbon price under different values of the subsidy parameter ϕ , positive jump probability β , and jump intensity κ are provided in Appendix E.

7. Conclusions

7.1. Concluding remarks

First, we propose a stochastic differential game modeling approach incorporating jump–diffusion uncertainties into a carbon-constrained profit maximization framework. We derive analytical solutions for the dynamic carbon prices, leveraging the Lévy process. To solve the coupled HJB equations, we construct an adjoint stochastic differential equation using the integral operator. We attain explicit solutions for the

government's optimal green subsidy, optimal carbon abatement inputs, and optimal production quantities for firms.

Second, we examine the interactions between the government and firms operating under carbon reduction constraints. The theoretical framework captures the dynamic evolution of optimal strategies among leaders and followers in a Stackelberg stochastic differential game. The stochastic optimization method focuses on asymmetric information, thus determining the dynamic interactions among multiple players in the presence of behavioral uncertainties. We introduce a dynamic payoff matrix that considers the potential outcomes of behavioral uncertainties in the extended model with principal-agent issues.

Third, we propose a methodology to assess the effectiveness of abatement subsidies. Carbon trading in our model offers a quantitative estimation of the social cost of carbon emissions based on optimal feedback strategies. Our analysis reveals that green subsidization is an effective carbon emission reduction policy, incentivizing carbon abatement inputs from high-emitting firms. Policy-oriented carbon trading harmonizes social welfare and profit maximization objectives, providing a feasible solution for incentive compatibility in carbon reduction and green transition. Our study underscores that the government necessitates to eliminate pretending behaviors through stringent regulatory measures. Green subsidization can effectively constrain total carbon emissions, which demonstrates the resilience of the policy-oriented carbon trading market.

7.2. Managerial implications

Our research has several managerial implications. The government plays a leadership role in carbon reduction by offering abatement subsidies, which incentivize enterprises to adopt green products and enhance green awareness. Introducing a precisely quantified carbon trading market provides an additional market-based mechanism that prompts businesses to invest more in green production lines, thereby significantly increasing the availability and share of green products in the marketplace. Firms in the carbon trading market have demonstrated a growing willingness to reduce their carbon emissions, as oriented by the carbon emission reduction policies. The enhanced reputation and market appeal of green products has led to an expansion in the market share of low-emission enterprises, which acts as a powerful incentive for high-emission firms to green manufacturing. The green transition guided by the government contributes to an overall improvement in social welfare, demonstrating the effectiveness of a combined approach involving government subsidies and a carbon trading market in driving sustainable development.

Additionally, the government motivates enterprises to progressively escalate their investments in emissions-cutting technical advancements by implementing abatement subsidies. The green transition in economic structure necessitates substantial additional investments propelled by the carbon emission reduction policies. These policies create a dynamic environment where the financial benefits of reducing emissions become tangible and lucrative. Confronted with the looming reality of missing government subsidies and incurring extra carbon emission costs for exceeding emission quotas, high-emitting firms display a heightened sense of urgency to curb their emissions. They are compelled to evolve into low-emission enterprises to evade potential penalties and capitalize on the burgeoning opportunities presented by the carbon trading market.

There are several possible extending directions in our game models. The non-cooperative stochastic differential game can be extended into cooperative versions in producing and carbon abatement. The cooperative interactions will lead to a more difficult formulation in the joint development of green technologies with a technological spillover rate. The multi-stage game extension can be constructed with accumulative outcomes in the finite horizon. The complicated information structure can be considered with dynamic probabilities of the regulated and abatement behaviors on incomplete information disclosure.

CRediT authorship contribution statement

Yuhan Li: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xiaoshan Su:** Validation, Methodology. **Jiandong Wang:** Validation, Methodology. **Haijun Yang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

This paper is no financial interest to report.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejor.2025.09.035>.

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